

Alex Jacobs

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Education

University of Waterloo

Sep. 2024 – Expected Jul. 2029

Candidate for Bachelor of Applied Science in Mechatronics Engineering

Waterloo, ON

Degree focus: Robotics and Automation, cGPA: 4.00/4.00

Experience

Research Assistant – Mobile Robotics & SLAM

Jan. 2025 – Apr. 2025

Sensing and Robotics for Infrastructure Lab, University of California, Los Angeles

Los Angeles, CA

- Developed a custom Gazebo simulator in C++ and Python to collect LiDAR, visual, and inertial sensor data from a quadrupedal robot performing waypoint navigation.
- Developed a metric evaluation framework in Python to benchmark state-of-the-art SLAM algorithms on open datasets.
- Debugged and tested proprietary laboratory software for robotic sensor data collection, processing, and visualization.
- Designed and 3D printed SolidWorks models to integrate new modules into existing ground and aquatic robot platforms.

Mechanical Designer – Mobile Robotics 🌐

Sep. 2023 – Jun. 2024

FIRST Robotics Competition Team, Northern Secondary School

Toronto, ON

- Modeled and manufactured a 3D-printed swerve drive system for our robot's chassis.
- Designed and assembled our robot's frame and drivetrain components.

Projects

Unmanned Surface Vehicle

Aug. 2023, Jan. 2025 – Apr. 2025

Sensing and Robotics for Infrastructure Lab, University of California, Los Angeles

Los Angeles, CA

- Maintained and created SolidWorks models of the platform's above- and below-water sensing kits.
- Modeled and 3D-printed a hydrodynamic fairing to reduce turbulence and drag around the vehicle's below-water sensing kit.
- Assisted with data collection using the platform's (detachable) above-water sensing kit for a visual-inertial navigation ablation study.

Forearm Guard 🌐

Aug. 2023

Personal Project

Toronto, ON

- Designed, modeled, and 3D printed a custom bite guard for therapists working with autistic children.

Costume and Prop Creation 🌐

Jun. 2022 - Present

405th: Halo Costume and Prop Maker Community

Toronto, ON

- Independently managed the full fabrication of life-size Halo props and costumes, including 3D-modeling, sizing, developing custom mounts, 3D-printing, painting, and assembling electronics.

Competitions

Skills Ontario & TDSB Mechanical CAD Competition

Mar. 5th, May 7th 2024

- Modeled mechanical components from technical drafts and physical parts using Fusion360.

Technical Skills

CAD Programs: SolidWorks, Onshape, Fusion 360, Blender | **Programming Languages:** C++, Python |

Manufacturing: 3D-Printing, Lathe, Milling Machine, Drill Press, Band Saw, Scroll Saw

Awards & Achievements

Term Distinction, University of Waterloo, Waterloo, ON.

Fall 2024

President's Scholarship of Distinction, University of Waterloo, Waterloo, ON.

May 2024